

Optimizing Credit Risk Assessment Models using Advanced Ensemble Machine Learning Techniques

Quantitative Research Division

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Abstract

Accurate credit risk assessment is critical for financial institution stability. Traditional models, such as Logistic Regression, often fail to capture complex, non-linear relationships within modern, high-dimensional financial datasets. This paper explores the efficacy of ensemble machine learning methods, specifically Random Forest and Extreme Gradient Boosting (XGBoost), in predicting loan default probabilities. By leveraging historical financial indicators, demographic data, and behavioral metrics, we demonstrate that ensemble architectures significantly outperform baseline models in both Area Under the Receiver Operating Characteristic Curve (ROC-AUC) and precision-recall metrics.

1 Introduction

The evaluation of creditworthiness is a fundamental operation in the banking sector. A fractional improvement in predicting the Probability of Default (PD) can save financial institutions millions of dollars and optimize capital allocation under Basel III frameworks. Recent advancements in Data Science and Machine Learning have shifted the paradigm from traditional statistical inference to high-capacity predictive modeling.

2 Methodology

2.1 Baseline: Logistic Regression

We define the probability of default $P(y = 1|x)$ using the standard logistic function:

$$P(y = 1|x) = \frac{1}{1 + e^{-(\beta_0 + \beta_1 x_1 + \dots + \beta_n x_n)}} \quad (1)$$

While highly interpretable, this model assumes linearity in the log-odds space, limiting its predictive boundary.

2.2 Ensemble Models

To capture non-linearities, we deploy tree-based ensemble methods.

2.2.1 Random Forest

Random Forest constructs a multitude of decision trees at training time. For classification, the output is the mode of the classes produced by individual trees. The Gini impurity for a node is calculated as:

$$G = 1 - \sum_{i=1}^C p_i^2 \quad (2)$$

where p_i is the probability of an item being classified for a distinct class.

2.2.2 XGBoost

XGBoost minimizes a regularized objective function combining a convex loss function and a penalty term for model complexity:

$$\mathcal{L}(\phi) = \sum_i l(y_i, \hat{y}_i) + \sum_k \Omega(f_k) \quad (3)$$

where $\Omega(f) = \gamma T + \frac{1}{2}\lambda\|w\|^2$. This strict regularization prevents overfitting on noisy financial data.

3 Results and Discussion

The models were trained on a proprietary dataset comprising 50,000 historical loan records. Hyperparameter tuning was conducted via grid search with 5-fold cross-validation. The XGBoost model achieved the highest ROC-AUC of 0.89, compared to 0.85 for Random Forest and 0.76 for Logistic Regression.

4 Conclusion

Ensemble machine learning models offer substantial improvements in credit risk analysis pipelines. Future work will investigate the integration of macroeconomic indicators and time-series modeling algorithms such as LSTM to capture temporal dynamics in default rates.